Fourier Series

Lecture One

Formula

$$\int x^n \, dx = \frac{x^{n+1}}{n+1}$$

$$\int uv \, dx = u \int v \, dx - \int \left[\frac{d}{dx}(u) \int v \, dx \right] dx$$

Advance :

$$\int uv \, dx = \pm \, u v_1 - u' v_2 + u'' v_3 - \ldots .$$

u = same
u' = Frist derivative
u'' = Second derivative
v_1 = Frist integration
v_2 = Second integration

Note : series will continue until derivative of u become 0.

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2+b^2} : \boxed{a \sin bx - b \cos bx}$$

$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2+b^2} : \boxed{a \cos bx + b \sin bx}$$

$\sin n\pi = 0$, $\sin 2n\pi = 0$,

$\cos n\pi = (-1)^n$

$\cos 2n\pi = (-1)^{2n} = 1$

Fourier Series Formula :

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \; ; \quad \alpha < x < \alpha + 2\pi$$

Then,

$$a_0 = \frac{1}{2\pi} \int_{\alpha}^{\alpha+2\pi} f(x)\, dx$$

$$a_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cos nx \; dx$$

$$b_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \sin nx \; dx \qquad \vdots$$

For : $-l < x < l$

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

where,

$$a_0 = \frac{1}{2l} \int_{-l}^{l} f(x)\, dx$$

$$a_n = \frac{1}{l} \int_{-l}^{l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$b_n = \frac{1}{l} \int_{-l}^{l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

Question 01 : Find fourier series to represent $f(x) = x^2$ in $(0, 2\pi)$ and hence deduce that

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

we know,

$$F(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos nx + b_n \sin nx \right]$$

$$\therefore \quad x^2 = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos nx + b_n \sin nx \right] \quad \text{— (1)}$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} x^2 \, dx$$

$$= \frac{1}{2\pi} \left[\frac{x^3}{3} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \times \frac{8\pi^3}{3}$$

$$= \frac{4\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx \, dx$$

$$= \frac{1}{\pi} \left[\frac{x^2 \sin nx}{n} - \frac{2x(-\cos nx)}{n^2} + \frac{2(-\sin nx)}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[4\pi^2 \times 0 + 4\pi . \frac{1}{n^2} \right]$$

$$= \frac{1}{\pi} \times \frac{4\pi}{n^2} = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx \, dx$$

$$= \frac{1}{\pi} \left[x^2 \left(\frac{-\cos nx}{n} \right) - \frac{2x(-\sin nx)}{n^2} + \frac{2\cos nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[-\frac{4\pi^2}{n} + \frac{2}{n^3} - \frac{2}{n^3} \right] = -\frac{4\pi}{n}$$

From (1), $x^2 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left[\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right]$

Putting $x = 0$ in (1), we get

$$0 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} (-1)^n \frac{4}{n^2}$$

$$\Rightarrow \quad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$

i.e., $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$

putting the value of a_0, a_n and b_n –

$$x^2 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left[\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right]$$

$$x^2 = \frac{4\pi^2}{3} + \left[\frac{4}{1^2} \cos x - \frac{4\pi}{1} \sin x + \frac{4}{2^2} \cos 2x - \frac{4\pi}{2} \sin 2x \right]$$

put $x = \pi$

$$\pi^2 = \frac{4\pi^2}{3} + \left[\frac{4}{1^2} (-1) + \frac{4}{2^2} (+1) + \dots \right]$$

$$\Rightarrow \quad \pi^2 = \frac{4\pi^2}{3} - \frac{4}{1^2} + \frac{4}{2^2} - \dots$$

$$\Rightarrow \quad \pi^2 - \frac{4\pi^2}{3} = -4 \left[\frac{1}{1^2} - \frac{1}{2^2} + \dots \right]$$

$$\Rightarrow \quad \frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \dots \qquad \text{[proved]}$$

lecture 02 :

Question 01 : Find fourier series of $f(x) = e^{ax}$ in $(0, 2\pi)$

we know,

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos nx + b_n \sin nx \right] \quad \text{--- (i)}$$

Here,

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x)\, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} e^{ax}\, dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{ax}}{a} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[\frac{e^{a(2\pi)} - 1}{a} \right]$$

$$= \frac{1}{2\pi} \left[\frac{e^{2\pi a} - 1}{a} \right]$$

$$= \frac{a}{2\pi(a^2)} \left[e^{2\pi a} - 1 \right]$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx\, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{ax} \cos nx\, dx$$

$$= \frac{1}{\pi} \left[\frac{e^{ax}}{a^2+n^2} (a \cos nx + n \sin nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{e^{a(2\pi)}(a \cdot 1 + n \cdot 0) - (a \cdot 1 + n \cdot 0)}{a^2+n^2} \right]$$

$$= \frac{1}{\pi} \left[\frac{e^{2\pi a} a - a}{a^2+n^2} \right]$$

$$= \frac{a}{\pi(a^2+n^2)} \left[e^{2\pi a} - 1 \right]$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx\, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{ax} \sin nx\, dx$$

$$= \frac{1}{\pi} \left[\frac{e^{ax}}{a^2+n^2} (a \sin nx - n \cos nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{e^{2\pi a}}{a^2+n^2} (a \sin n2\pi - n \cos n2\pi) - \frac{1}{a^2+n^2} (a \sin 0 - n \cos 0) \right]$$

$$= \frac{1}{\pi} \left[\frac{e^{2\pi a}}{a^2+n^2} (a \cdot 0 - n \cdot 1) - \frac{1}{a^2+n^2} (a \cdot 0 - n \cdot 1) \right]$$

$$= \frac{1}{\pi} \left[\frac{-e^{2\pi a} n}{a^2+n^2} + \frac{n}{a^2+n^2} \right]$$

$$= \frac{n}{\pi(a^2+n^2)} \left[1 - e^{2\pi a} \right]$$

putting the value of a_0, a_n and b_n –

$$e^{ax} = \frac{1}{2\pi}\left[\frac{e^{2\pi a}-1}{a}\right] + \sum_{n=1}^{\infty}\left[\frac{a}{\pi(a^2+n^2)}\left(e^{2\pi a}-1\right)\cos nx + \frac{n}{\pi(a^2+n^2)}\left(1-e^{2\pi a}\right)\sin nx\right]$$

Question 02 : Find the fourier series to represents the function if

$$f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

we know, $f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos nx + b_n \sin nx\right]$

Here,

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{2\pi}\left[\int_{-\pi}^{0}(-\pi)\,dx + \int_{0}^{\pi} x\,dx\right]$$

$$= \frac{1}{2\pi}\left[-\pi(x)\Big|_{-\pi}^{0} + \left(\frac{x^2}{2}\right)\Big|_{0}^{\pi}\right]$$

$$= \frac{1}{2\pi}\left[-\pi^2 + \frac{\pi^2}{2}\right]$$

$$= -\frac{\pi}{4}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} -\pi\cos nx\,dx + \int_{0}^{\pi} x\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[-\pi\left(\frac{\sin nx}{n}\right)\Big|_{-\pi}^{0} + \left(\frac{x\sin nx}{n}\right)\Big|_{0}^{\pi} + \left(\frac{\cos nx}{n^2}\right)\Big|_{0}^{\pi}\right]$$

$$= -\frac{1}{\pi}\times\pi\left[\frac{\sin n0}{n} - \frac{\sin(-n\pi)}{n}\right] + \frac{1}{\pi}\left[\frac{\pi\sin n\pi}{n} - 0\right] + \left[\frac{\cos n\pi}{n^2} - \frac{\cos 0}{n^2}\right]$$

$$= 0 + \frac{1}{\pi}\left[\frac{(-1)^n}{n^2} - \frac{1}{n^2}\right]$$

$$= \frac{1}{\pi n^2}\left[(-1)^n - 1\right]$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} -\pi\sin nx\,dx + \int_{0}^{\pi} x\sin nx\,dx\right]$$

$$= \frac{1}{\pi}\left[-\pi\left(\frac{-\cos nx}{n}\right)\Big|_{-\pi}^{0} + \left(-\frac{x\cos nx}{n}\right)\Big|_{0}^{\pi} + \left(\frac{\sin nx}{n^2}\right)\Big|_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[-\pi\left(\frac{-\cos n0}{n} - \frac{-\cos(-n\pi)}{n}\right) - \left(\frac{\pi\cos n\pi}{n} - 0\right) + \left(\frac{\sin n\pi}{n^2} - \frac{\sin 0}{n^2}\right)\right]$$

$$= \frac{1}{\pi}\left[-\pi\left(\frac{-1-(-1)^n}{n}\right) - \frac{\pi(-1)^n}{n} + 0\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi}{n}\left(1-(-1)^n\right) - \frac{\pi(-1)^n}{n}\right]$$

$$= \frac{1}{n}\left[1-(-1)^n-(-1)^n\right] = \frac{n}{\pi}\left[1-(-1)^n\right]$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}(-\pi\sin nx\,dx) + \int_{0}^{\pi} x\sin nx\,dx\right]$$

$$= -\left[\frac{-\cos nx}{n}\right]_{-\pi}^{0} + \frac{1}{\pi}\left[x\left(\frac{-\cos nx}{n}\right) + \frac{\sin nx}{n^2}\right]_{0}^{\pi}$$

$$= \left[\frac{1}{n} - \frac{(-1)^n}{n}\right] + \frac{1}{\pi}\left[-\pi\frac{(-1)^n}{n}\right]$$

$$= \frac{1}{n} - \frac{(-1)^n}{n} - \frac{(-1)^n}{n} = \frac{1-2(-1)^n}{n}$$

$$f(x) = -\frac{\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{1}{n\pi^2}(-1)^{n2-1}\cos nx + \left(\frac{1-2(-1)^n}{n}\right)\sin nx\right]$$

Lecture 03

Even and odd concept

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n\cos nx + b_n\sin nx\right] \qquad -\pi < x < \pi$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

Even function ⊕

$$F(x) = F(-x)$$

$\cos x$, x^2, x^4, x^6, $x \sin x$

⊖ ⊖

⊖ Odd function

$$f(x) = -f(-x)$$

$\sin x$, x , x^3, $x \cos x$

− ⊕

For fourier series :

If $f(x)$ Even

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x)\, dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx\, dx$$

$$b_n = 0$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx)$$

If $f(x)$ Odd

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx\, dx$$

$$f(x) = \sum_{n=1}^{\infty} (b_n \sin nx)$$

Lecture 03

Even and odd concept

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos nx + b_n \sin nx \right] \qquad -\pi < x < \pi$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx$$

Lecture 04 : Even :

Question 01 : *Find the fourier series for* $f(x) = x \sin x$, $-\pi < x < \pi$ *and proof* $\frac{\pi}{4} = \frac{1}{2} - \frac{1}{1\times 3} - \frac{1}{3\times 5} + \dots$

Solution : *As even,*

$$b_n = 0$$

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x\sin x\,dx$$

$$= \frac{1}{\pi}\left[x(-\cos x) - \sin x\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[-\pi(-1) - 0\right] = 1$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\sin x\cos nx\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x\left\{\sin(1+n)x + \sin(1-n)x\right\}dx$$

$$= \frac{1}{\pi}\left[x\left(-\frac{\cos(1+n)x}{1+n}\right) + \frac{\sin(1+n)x}{1+n}\right]_0^{\pi}$$

$$+ x\left[x\left(-\frac{\cos(1-n)x}{1-n}\right) + \frac{\sin(1-n)x}{1-n}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left\{\pi\left(-\frac{\cos(1+n)\pi}{1+n}\right) - \frac{\cos(1-n)\pi}{1-n}\right\}$$

$$= \frac{1}{\pi}\left[\frac{(-1)^n}{1+n} + \frac{(-1)^n}{1-n}\right]$$

$$= \frac{2(-1)^n}{1-n^2}, \quad n \neq 1$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\sin x\cos nx\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x\left(\sin(1+n)x + \sin(1-n)x\right)dx$$

$$= \frac{1}{\pi}\left[x\left(-\frac{\cos(1+n)x}{1+n}\right) + \frac{\sin(1+n)x}{1+n}\right]_0^{\pi}$$

$$+ x\left(-\frac{\cos(1-n)x}{1-n}\right) + \frac{\sin(1-n)x}{1-n}\Bigg]_0^{\pi}$$

$$= \frac{1}{\pi}\left\{\pi\left(-\frac{\cos(1+n)\pi}{1+n}\right) - \frac{\cos(1-n)\pi}{1-n}\right\}$$

$$= \frac{1}{\pi}\left[\frac{(-1)^n}{1+n} + \frac{(-1)^n}{1-n}\right]$$

$$= \frac{2(-1)^n}{1-n^2}, \quad n \neq 1$$

$$\therefore\ f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx = 1 + \sum_{n=1}^{\infty} \frac{2(-1)^n}{1-n^2}\cos nx, \quad -\pi < x < \pi$$

Put $x = 0$ *in* $f(x)$, *we get*

$$0 = 1 + \sum_{n=1}^{\infty}\frac{2(-1)^n}{1-n^2} \Rightarrow \sum_{n=1}^{\infty}\frac{(-1)^n}{1-n^2} = -\frac{1}{2} \Rightarrow \sum_{n=1}^{\infty}\frac{1}{n^2-1} = -\frac{1}{2} \Rightarrow \sum_{n=1}^{\infty}\frac{1}{n^2-1} = \sum_{n=1}^{\infty}\frac{1}{(2n-1)(2n+1)}$$

$$= \frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right) = \frac{1}{2}\left(1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \dots\right) = \frac{1}{2}\left(1 - \frac{1}{3\times 5} + \frac{1}{3\times 5} - \frac{1}{5\times 7} + \dots\right)$$

$$= \frac{1}{2} - \frac{1}{1\times 3} + \frac{1}{3\times 5} - \frac{1}{5\times 7} + \dots \Rightarrow \frac{\pi}{4} = \frac{1}{2} - \frac{1}{1\times 3} + \frac{1}{3\times 5} - \frac{1}{5\times 7} + \dots$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$= a_0 + a_1 \cos x + \sum_{n=2}^{\infty} a_n \cos nx$$

$$= 1 - \frac{1}{2}\cos x + \sum_{n=2}^{\infty} \frac{2(-1)^n}{1-n^2} \cos nx .$$

$$\therefore \quad x \sin x = 1 - \frac{1}{2}\cos x + 2\left[\frac{+1}{-3}\cos 2x + \frac{\cos 3x}{8} + \dots\right]$$

$$\Rightarrow x \sin x = 1 - \frac{\cos x}{2} - \frac{2}{3}\cos 2x + \frac{\cos 3x}{4} + \dots$$

putting $n = \frac{\pi}{2}$

$$\Rightarrow \frac{\pi}{2} = 1 - 0 - \frac{2}{3}(-1) + \frac{1}{4} \times 0 + \dots$$

$$\Rightarrow \frac{\pi}{2} = 1 + \frac{2}{3} + \dots$$

$$\therefore \frac{\pi}{4} = \frac{1}{2} + \frac{1}{3.1} + \dots$$

Question 02 :

Find fourier series of $f(x) = x^2, \quad -\pi < x < \pi$ *and proof.*

(i) $$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

As even,

$$b_n = 0$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 \, dx$$

$$= \frac{\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx \, dx$$

$$= \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{2x \cos nx}{n^2} + \frac{\sin nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[2\pi \times \frac{(-1)^n}{n^2} \right]$$

$$= \frac{4(-1)^n}{n^2}$$

$$x^2 = \frac{\pi^2}{3} + 4 \left[-\frac{\cos x}{1} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \dots \right]$$

put $x = \pi$

$$\pi^2 = \frac{\pi^2}{3} + 4 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right)$$

$$\frac{2\pi^2}{3} = 4 \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right)$$

$$\therefore \frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \qquad [\textit{proved}]$$

put $x = 0$

$$0 = \frac{\pi^2}{3} + 4 \left(-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots \right)$$

$$\therefore \frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots \qquad [\textit{proved}]$$

Lecture 05 (odd)

Question 01 : Find fourier series for $f(x) = x \cos x$

$-\pi < x < \pi$

Solution : As odd function

so, $a_0 = a_n = 0$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos x \sin nx \, dx$$

$$= \frac{2}{\pi} \times \frac{1}{2} \int_0^{\pi} x \left[\sin(1+n)x - x \sin(1-n)x \right] dx$$

$$= \frac{1}{\pi} \left[x \left\{ \frac{-\cos(1+n)x}{1+n} + \frac{\sin(1+n)x}{(1+n)^2} + \frac{x \cos(1-n)x}{1-n} - \frac{\sin(1-n)x}{(1-n)^2} \right]_0^{\pi} \right.$$

$$= \frac{1}{\pi} \left[\pi \frac{(-\cos(1+n)\pi)}{1+n} + \frac{\pi \cos(1-n)\pi}{1-n} \right]$$

$$= \left(+ \frac{(-1)^n}{1+n} - \frac{(-1)^n}{1-n} \right)$$

$$= -\frac{2(-1)^n \times n}{1-n^2} = \frac{2n(-1)^n}{n^2-1}, \quad n \neq 1$$

$$f(x) = b_1 \sin x + \sum_{n=2}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin x \cos nx \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin 2x \, dx$$

$$= \frac{1}{\pi} \left[x \left(-\frac{\cos 2\pi}{2} \right) + \frac{\sin 2x}{4} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\pi \left(-\frac{\cos 2\pi}{2} \right) + \frac{\sin 2\pi}{4} - 0 \right]$$

$$= \frac{1}{\pi} \left[-\frac{\pi(-1)}{2} \right] = \frac{(-1)^n}{2} = -\frac{1}{2}$$

$$f(x) = -\frac{1}{2} \sin x + \sum_{n=2}^{\infty} \frac{2(-1)^n n}{n^2 - 1} \sin nx$$

Question 2: *Find fourier series of* $f(x) = x \; ; \quad -\pi < x < \pi$

As odd function.

$$a_0 = a_n = 0$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin nx \, dx$$

$$= \frac{2}{\pi} \left[x \left(-\frac{\cos nx}{n} \right) + \frac{\sin nx}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\pi \left(-\frac{\cos n\pi}{n} \right) + \frac{\sin n\pi}{n^2} - 0 \right]$$

$$= \frac{2}{\pi} \left[\pi \left(-\frac{(-1)^n}{n} \right) \right]$$

$$= \frac{2}{\pi} \left[\pi \times \frac{(-1)^{n-1}}{n} \right]$$

$$= -\frac{2(-1)^n}{n}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{-2(-1)^{n-1}}{n} \sin nx$$

Lecture - 06 :

Half wave cosine series

Half wave cosine series on interval $0 \le x \le l$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l}\int_0^l f(x)\,dx$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

If $l = \pi$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx.$$

Question 01 : Express $f(x) = x$ as a half range cosine series in interval $0 < x < 2$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{2}\right)$$

$$a_0 = \frac{1}{2}\int_0^2 f(x)\,dx = \frac{1}{2}\int_0^2 x\,dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^2 = 1$$

$$a_n = \frac{2}{2}\int_0^2 f(x)\cos\left(\frac{n\pi x}{2}\right)dx = \int_0^2 x\cos\left(\frac{n\pi x}{2}\right)dx$$

$$= \left[x\,\frac{\sin\left(\frac{n\pi x}{2}\right)}{\frac{n\pi}{2}} + \frac{\cos\left(\frac{n\pi x}{2}\right)}{\left(\frac{n\pi}{2}\right)^2}\right]_0^2 = \frac{4(-1)^n}{n^2\pi^2} - \frac{4}{n^2\pi^2}$$

$$f(x) = x = 1 + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \left[(-1)^n - 1\right] \cos\left(\frac{n\pi x}{2}\right)$$

Question 02 : Obtain half range cosine series for

$$f(x) = \begin{cases} kx, & 0 \le x \le \frac{l}{2} \\ k(l-x), & \frac{l}{2} \le x \le l \end{cases}$$

and proof. $\frac{1}{1^r} + \frac{1}{3^r} + \frac{1}{5^r} + \cdots = \frac{\pi^2}{8}$

Solution : $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$

$$a_0 = \frac{1}{l}\left[\int_0^{l/2} kx\, dx + \int_{l/2}^{l} k(l-x)\, dx\right]$$

$$= \frac{1}{l}\left[\left(\frac{kx^2}{2}\right)_0^{l/2} + \frac{k}{l}\left[lx - \frac{x^2}{2}\right]_{l/2}^{l}\right]$$

$$= \frac{k}{l}\left(\frac{l^2}{8}\right) + \frac{k}{l}\left[l^2 - \frac{l^2}{2} - \frac{l^2}{2} + \frac{l^2}{8}\right]$$

$$= \frac{kl}{8} + \frac{kl}{l}\left[\frac{l^2}{8}\right]$$

$$= \frac{kl}{8} + \frac{kl}{8}$$

$$= \frac{kl}{4}$$

$$a_n = \frac{2}{l}\left[\int_0^{l/2} kx \cos\left(\frac{n\pi x}{l}\right) dx + \int_{l/2}^{l} k(l-x) \cos\left(\frac{n\pi x}{l}\right) dx\right]$$

$$= \frac{2}{l}\int_0^{l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

For $r = 2$, putting into the series expansion and comparing coefficients, we get

$$\frac{1}{1^r} + \frac{1}{3^r} + \frac{1}{5^r} + \cdots = \frac{\pi^2}{8}$$

$$a_n = \frac{2}{l}\left[\int_0^{l/2} kx\left(\frac{\cos n\pi x}{l}\right)dx + \int_{l/2}^{l} k(l-x)\cos\left(\frac{n\pi x}{l}\right)dx\right]$$

$$= \frac{2k}{l}\left[x\,\frac{\left(\sin\left(\frac{n\pi x}{l}\right)\right)}{\frac{n\pi}{l}} + \frac{\cos\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)^2}\right]_0^{\frac{l}{2}} + \frac{2k}{l}\left[(l-x)\,\frac{\left(\sin\left(\frac{n\pi x}{l}\right)\right)}{\frac{n\pi}{l}} - \frac{\cos\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)^2}\right]_{l/2}^{l}$$

$$= \frac{2k}{l}\left[\frac{l}{2}\,\frac{\sin\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{l}} + \frac{\cos\left(\frac{n\pi}{2}\right)}{\frac{n^2\pi^2}{l^2}} - \frac{1}{\frac{n^2\pi^2}{l^2}}\right] + \frac{2k}{l}\left[0 - \frac{(-1)^n}{\frac{n^2\pi^2}{l^2}} - \frac{l}{2}\,\frac{\sin\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{l}}\right]$$

$$= \frac{2k}{l}\left[\frac{l^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{l^2}{n^2\pi^2}\cos\left(\frac{n\pi}{2}\right) - \frac{l^2}{n^2\pi^2}\right] + \frac{2k}{l}\left[-\frac{(-1)^n l^2}{n^2\pi^2} - \frac{l^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{\cos\left(\frac{n\pi}{2}\right)l^2}{n^2\pi^2}\right]$$

$$a_n = \frac{2k}{l}\left[\frac{l^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{l^2}{n^2\pi^2}\cos\left(\frac{n\pi}{2}\right) - \frac{l^2}{n^2\pi^2} - \frac{(-1)^n l^2}{n^2\pi^2} - \frac{l^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{l^2}{n^2\pi^2}\cos\left(\frac{n\pi}{2}\right)\right]$$

$$\therefore\quad a_n = \frac{2k}{l}\left[\frac{2l^2}{n^2\pi^2}\cos\left(\frac{n\pi}{2}\right) - \frac{l^2}{n^2\pi^2} - \frac{(-1)^n l^2}{n^2\pi^2}\right]$$

$$\therefore\quad a_n = \frac{2k\cdot l}{l\,n^2\pi^2}\left[2\cos\left(\frac{n\pi}{2}\right) - 1 - (-1)^n\right]$$

$$= \frac{2kl}{n^2\pi^2}\left[2\cos\left(\frac{n\pi}{2}\right) - 1 - (-1)^n\right]$$

$a_1 = 0$

$a_2 = \dfrac{kl}{2\pi^2}(-2-1-1) = \dfrac{-2kl}{\pi^2}$

$= \dfrac{-4kl}{2\pi^2} = \dfrac{-8kl}{2^2\pi^2}$ *(for easily solving)*

$a_3 = 0$

$a_4 = 0$

$a_5 = 0$

$a_6 = \dfrac{-8kl}{6^2\pi^2}$

$$f(x) = a_0 + a_1 \cos\left(\frac{\pi x}{\ell}\right) + a_2 \cos\left(\frac{2\pi x}{\ell}\right) + a_3 \cos\left(\frac{3\pi x}{\ell}\right)$$

$$= \frac{k\ell}{4} - \frac{8k\ell}{2^2\pi^2} \cos\left(\frac{2\pi x}{\ell}\right) - \frac{8k\ell}{3^2\pi^2} \cos\left(\frac{3\pi x}{\ell}\right) + \cdots$$

$x = 0$

$$0 = \frac{k\ell}{4} - \frac{8k\ell}{2^2\pi^2} - \frac{8k\ell}{3^2\pi^2} + \cdots$$

$$\Rightarrow \frac{k\ell}{4} \times \frac{2^2\pi^2}{8k\ell} = \left(1 + \frac{1}{3^2} + \cdots\right)$$

$$\therefore \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \cdots \qquad [proved]$$

Lecture - 07

Half wave sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{\ell}\right)$$

$$b_n = \frac{2}{\ell} \int_0^{\ell} f(x) \sin\left(\frac{n\pi x}{\ell}\right) dx$$

If $\ell = \pi$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

Question 01 : *Find half range sine series for* $f(x) = 2x - 1$, $0 < x < 1$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(n\pi x)$$

$$b_n = \frac{2}{1} \int_0^{1} f(x) \sin(n\pi x) \, dx$$

$$= 2 \int_0^{1} (2x-1) \sin(n\pi x) \, dx$$

$$= 2 \left[(2x-1) \left(- \frac{\cos(n\pi x)}{n\pi} \right) + \frac{\sin(n\pi x)}{n^2\pi^2} \right]_0^1$$

$$= 2 \left[\frac{-(-1)^n}{n\pi} + \frac{-1}{n\pi} \right] = \frac{-2}{n\pi} \left[1 + (-1)^n \right]$$

$$\therefore \quad f(x) = \sum_{n=1}^{\infty} \frac{-2}{n\pi} \left[1 + (-1)^n \right] \sin(n\pi x)$$

Question 02 : Express $f(x) = x$ as a half range sine series in $0 < x < 2$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$$

$$b_n = \frac{2}{2}\int_0^2 f(x) \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= \int_0^2 x \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= \left[x\left(\frac{-\cos\left(\frac{n\pi x}{2}\right)}{\frac{n\pi}{2}}\right) + \frac{\sin\left(\frac{n\pi x}{2}\right)}{\left(\frac{n\pi}{2}\right)^2} \right]_0^2$$

$$= \left[2\left(\frac{-\cos\left(\frac{n\pi.2}{2}\right)}{\frac{n\pi}{2}}\right) + \frac{\sin\left(\frac{n\pi.2}{2}\right)}{\left(\frac{n\pi}{2}\right)^2} \right] - 0$$

$$= \left[2\left(\frac{-\cos(n\pi)}{\frac{n\pi}{2}}\right) + \frac{\sin(n\pi)}{\left(\frac{n\pi}{2}\right)^2} \right] = \frac{-4\cos(n\pi)}{n\pi}$$

$$= \frac{-4(-1)^n}{n\pi}$$

$$\therefore f(x) = -\frac{4}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin\left(\frac{n\pi x}{2}\right)$$

$$= -\frac{4}{\pi}\left[-\sin\left(\frac{\pi x}{2}\right) + \frac{1}{2}\sin\left(\frac{2\pi x}{2}\right) - \frac{1}{3}\sin\left(\frac{3\pi x}{2}\right) + \cdots \right]$$